PG No. 45 $C_{6h}(c)$ $6/m$ [hexagonal] (internal axial octupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(3,3)}[g](A_u)$

** symmetry

1

** expression

$$\frac{\sqrt{70}G_{3c}x\left(x^2-3y^2\right)}{28}+\frac{\sqrt{70}G_{3s}y\left(3x^2-y^2\right)}{28}-\frac{\sqrt{42}G_{3x}x\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{42}G_{3y}y\left(x^2+y^2-4z^2\right)}{28}$$
$$+\frac{\sqrt{105}G_{xyz}xyz}{7}-\frac{\sqrt{7}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}G_z^{\beta}z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(3,1)}[g](A_g)$

** symmetry

z

** expression

$$\frac{2\sqrt{210}G_{3x}xz}{35}+\frac{2\sqrt{210}G_{3y}yz}{35}+\frac{\sqrt{21}G_{xyz}xy}{7}-\frac{3\sqrt{35}G_z^{\alpha}\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}G_z^{\beta}\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](A_g)$

** symmetry

z

** expression

$$\frac{\sqrt{210}G_{3c}xz\left(x^2-3y^2\right)}{24}+\frac{\sqrt{210}G_{3s}yz\left(3x^2-y^2\right)}{24}-\frac{5\sqrt{14}G_{3x}xz\left(3x^2+3y^2-4z^2\right)}{56}-\frac{5\sqrt{14}G_{3y}yz\left(3x^2+3y^2-4z^2\right)}{56}$$
$$-\frac{\sqrt{35}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{\sqrt{21}G_z^{\alpha}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

$\vec{\mathbb{G}}_1^{(3,1)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{3\sqrt{7}G_{3c}\left(x-iy\right)^2}{28}+\frac{3\sqrt{7}iG_{3s}\left(x-iy\right)^2}{28}-\frac{\sqrt{105}G_{3x}\left(3x^2+2ixy+y^2-4z^2\right)}{140}$$
$$-\frac{\sqrt{105}iG_{3y}\left(x^2-2ixy+3y^2-4z^2\right)}{140}+\frac{\sqrt{42}iG_{xyz}z\left(x-iy\right)}{14}-\frac{3\sqrt{70}G_z^{\alpha}z\left(x+iy\right)}{70}+\frac{\sqrt{42}G_z^{\beta}z\left(x-iy\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{105}G_{3c}\left(4x^4+13ix^3y-21x^2y^2-3x^2z^2-15ixy^3+6ixyz^2+3y^4+3y^2z^2\right)}{168}$$
$$-\frac{\sqrt{105}iG_{3s}\left(3x^4+15ix^3y-21x^2y^2+3x^2z^2-13ixy^3-6ixyz^2+4y^4-3y^2z^2\right)}{168}$$
$$-\frac{\sqrt{7}G_{3x}\left(4x^4+5ix^3y+3x^2y^2-27x^2z^2+5ixy^3-30ixyz^2-y^4+3y^2z^2+4z^4\right)}{56}$$
$$+\frac{\sqrt{7}iG_{3y}\left(x^4+5ix^3y-3x^2y^2-3x^2z^2+5ixy^3-30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{56}-\frac{\sqrt{70}iG_{xyz}z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{28}$$
$$-\frac{5\sqrt{42}G_z^{\alpha}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}G_z^{\beta}z\left(5x^3+9ix^2y-9xy^2-2xz^2-5iy^3+2iyz^2\right)}{56}$$

$\vec{\mathbb{G}}_1^{(3,1)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{3\sqrt{7}G_{3c}\left(x+iy\right)^2}{28}-\frac{3\sqrt{7}iG_{3s}\left(x+iy\right)^2}{28}-\frac{\sqrt{105}G_{3x}\left(3x^2-2ixy+y^2-4z^2\right)}{140}$$
$$+\frac{\sqrt{105}iG_{3y}\left(x^2+2ixy+3y^2-4z^2\right)}{140}-\frac{\sqrt{42}iG_{xyz}z\left(x+iy\right)}{14}-\frac{3\sqrt{70}G_z^{\alpha}z\left(x-iy\right)}{70}+\frac{\sqrt{42}G_z^{\beta}z\left(x+iy\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{105}G_{3c}\left(4x^4 - 13ix^3y - 21x^2y^2 - 3x^2z^2 + 15ixy^3 - 6ixyz^2 + 3y^4 + 3y^2z^2\right)}{168} + \frac{\sqrt{105}iG_{3s}\left(3x^4 - 15ix^3y - 21x^2y^2 + 3x^2z^2 + 13ixy^3 + 6ixyz^2 + 4y^4 - 3y^2z^2\right)}{168} - \frac{\sqrt{7}G_{3x}\left(4x^4 - 5ix^3y + 3x^2y^2 - 27x^2z^2 - 5ixy^3 + 30ixyz^2 - y^4 + 3y^2z^2 + 4z^4\right)}{56} - \frac{\sqrt{7}iG_{3y}\left(x^4 - 5ix^3y - 3x^2y^2 - 3x^2z^2 - 5ixy^3 + 30ixyz^2 - 4y^4 + 27y^2z^2 - 4z^4\right)}{56} + \frac{\sqrt{70}iG_{xyz}z\left(x^3 - 6ix^2y - 6xy^2 + xz^2 + iy^3 + iyz^2\right)}{28} - \frac{5\sqrt{42}G_z^{\alpha}z\left(x - iy\right)\left(3x^2 + 3y^2 - 4z^2\right)}{168} + \frac{\sqrt{70}G_z^{\beta}z\left(5x^3 - 9ix^2y - 9xy^2 - 2xz^2 + 5iy^3 - 2iyz^2\right)}{56}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](A_u)$

** symmetry

z

** expression

$$\frac{3\sqrt{70}G_{3c}y\left(3x^2 - y^2\right)}{56} - \frac{3\sqrt{70}G_{3s}x\left(x^2 - 3y^2\right)}{56} - \frac{\sqrt{42}G_{3x}y\left(x^2 + y^2 - 4z^2\right)}{56} + \frac{\sqrt{42}G_{3y}x\left(x^2 + y^2 - 4z^2\right)}{56} - \frac{\sqrt{105}G_{xyz}z\left(x - y\right)\left(x + y\right)}{14} + \frac{\sqrt{105}G_z^{\beta}xyz}{7}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{35}iG_{3c}z\left(x - iy\right)^2}{56} + \frac{3\sqrt{35}G_{3s}z\left(x - iy\right)^2}{56} + \frac{\sqrt{21}iG_{3x}z\left(11x^2 + 10ixy + y^2 - 4z^2\right)}{56} - \frac{\sqrt{21}G_{3y}z\left(x^2 - 10ixy + 11y^2 - 4z^2\right)}{56} + \frac{\sqrt{210}iG_{xyz}\left(x^2y + ixy^2 - ixz^2 - yz^2\right)}{28} - \frac{3\sqrt{14}iG_z^{\alpha}\left(x + iy\right)\left(x^2 + y^2 - 4z^2\right)}{56} + \frac{\sqrt{210}iG_z^{\beta}\left(x^3 + ix^2y - xy^2 - 2xz^2 - iy^3 + 2iyz^2\right)}{56}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](E_{1u}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{3\sqrt{35}iG_{3c}z\left(x + iy\right)^2}{56} + \frac{3\sqrt{35}G_{3s}z\left(x + iy\right)^2}{56} - \frac{\sqrt{21}iG_{3x}z\left(11x^2 - 10ixy + y^2 - 4z^2\right)}{56} - \frac{\sqrt{21}G_{3y}z\left(x^2 + 10ixy + 11y^2 - 4z^2\right)}{56} - \frac{\sqrt{210}iG_{xyz}\left(x^2y - ixy^2 + ixz^2 - yz^2\right)}{28} + \frac{3\sqrt{14}iG_z^{\alpha}\left(x - iy\right)\left(x^2 + y^2 - 4z^2\right)}{56} - \frac{\sqrt{210}iG_z^{\beta}\left(x^3 - ix^2y - xy^2 - 2xz^2 + iy^3 - 2iyz^2\right)}{56}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(3,0)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{21}G_{3x}yz}{7} - \frac{\sqrt{21}G_{3y}xz}{7} - \frac{\sqrt{210}G_{xyz}\left(x - y\right)\left(x + y\right)}{28} + \frac{\sqrt{210}G_z^{\beta}xy}{14}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{70}G_{3c}yz\left(3x^2 - y^2\right)}{8} - \frac{\sqrt{70}G_{3s}xz\left(x^2 - 3y^2\right)}{8} - \frac{5\sqrt{42}G_{3x}yz\left(3x^2 + 3y^2 - 4z^2\right)}{168} + \frac{5\sqrt{42}G_{3y}xz\left(3x^2 + 3y^2 - 4z^2\right)}{168} + \frac{\sqrt{105}G_{xyz}\left(x - y\right)\left(x + y\right)\left(x^2 + y^2 - 6z^2\right)}{42} - \frac{\sqrt{105}G_z^{\beta}xy\left(x^2 + y^2 - 6z^2\right)}{21}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{210}iG_{3c}\left(x-iy\right)^2}{56}-\frac{\sqrt{210}G_{3s}\left(x-iy\right)^2}{56}+\frac{\sqrt{14}iG_{3x}\left(5x^2+6ixy-y^2-4z^2\right)}{56}+\frac{\sqrt{14}G_{3y}\left(x^2+6ixy-5y^2+4z^2\right)}{56}+\frac{\sqrt{21}iG_z^{\alpha}z\left(x+iy\right)}{7}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\begin{aligned}&-\frac{\sqrt{105}iG_{3c}\left(2x^4+17ix^3y-21x^2y^2+9x^2z^2-11ixy^3-18ixyz^2+5y^4-9y^2z^2\right)}{168}\\&-\frac{\sqrt{105}G_{3s}\left(5x^4+11ix^3y-21x^2y^2-9x^2z^2-17ixy^3+18ixyz^2+2y^4+9y^2z^2\right)}{168}\\&-\frac{5\sqrt{7}iG_{3x}\left(2x^4+ix^3y+3x^2y^2-15x^2z^2+ixy^3-6ixyz^2+y^4-9y^2z^2+4z^4\right)}{168}\\&+\frac{5\sqrt{7}G_{3y}\left(x^4-ix^3y+3x^2y^2-9x^2z^2-ixy^3+6ixyz^2+2y^4-15y^2z^2+4z^4\right)}{168}-\frac{\sqrt{70}G_{xyz}z\left(x^3-xz^2-iy^3+iyz^2\right)}{12}\\&-\frac{5\sqrt{42}iG_z^{\alpha}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{70}iG_z^{\beta}z\left(x^3-3ix^2y+3xy^2-2xz^2-iy^3+2iyz^2\right)}{24}\end{aligned}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$-\frac{\sqrt{210}iG_{3c}\left(x+iy\right)^2}{56}-\frac{\sqrt{210}G_{3s}\left(x+iy\right)^2}{56}-\frac{\sqrt{14}iG_{3x}\left(5x^2-6ixy-y^2-4z^2\right)}{56}+\frac{\sqrt{14}G_{3y}\left(x^2-6ixy-5y^2+4z^2\right)}{56}-\frac{\sqrt{21}iG_z^{\alpha}z\left(x-iy\right)}{7}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\begin{aligned}&\frac{\sqrt{105}iG_{3c}\left(2x^4-17ix^3y-21x^2y^2+9x^2z^2+11ixy^3+18ixyz^2+5y^4-9y^2z^2\right)}{168}\\&-\frac{\sqrt{105}G_{3s}\left(5x^4-11ix^3y-21x^2y^2-9x^2z^2+17ixy^3-18ixyz^2+2y^4+9y^2z^2\right)}{168}\\&+\frac{5\sqrt{7}iG_{3x}\left(2x^4-ix^3y+3x^2y^2-15x^2z^2-ixy^3+6ixyz^2+y^4-9y^2z^2+4z^4\right)}{168}\\&+\frac{5\sqrt{7}G_{3y}\left(x^4+ix^3y+3x^2y^2-9x^2z^2+ixy^3-6ixyz^2+2y^4-15y^2z^2+4z^4\right)}{168}-\frac{\sqrt{70}G_{xyz}z\left(x^3-xz^2+iy^3-iyz^2\right)}{12}\\&+\frac{5\sqrt{42}iG_z^{\alpha}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{168}-\frac{\sqrt{70}iG_z^{\beta}z\left(x^3+3ix^2y+3xy^2-2xz^2+iy^3-2iyz^2\right)}{24}\end{aligned}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\begin{aligned}&\frac{\sqrt{210}iG_{3c}z\left(x+iy\right)}{28}+\frac{\sqrt{210}G_{3s}z\left(x+iy\right)}{28}-\frac{3\sqrt{14}iG_{3x}z\left(x-iy\right)}{28}-\frac{3\sqrt{14}G_{3y}z\left(x-iy\right)}{28}\\&-\frac{\sqrt{35}G_{xyz}\left(x^2+y^2-2z^2\right)}{28}+\frac{\sqrt{21}iG_z^{\alpha}\left(x-iy\right)^2}{28}-\frac{\sqrt{35}iG_z^{\beta}\left(x^2+y^2-2z^2\right)}{28}\end{aligned}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{105}iG_{3c}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{105}G_{3s}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{168}-\frac{5\sqrt{7}iG_{3x}z\left(x-iy\right)\left(13x^2-14ixy-y^2-8z^2\right)}{168}$$
$$+\frac{5\sqrt{7}G_{3y}z\left(x-iy\right)\left(x^2-14ixy-13y^2+8z^2\right)}{168}+\frac{\sqrt{70}G_{xyz}\left(x^4-7ix^3y-12x^2y^2+6x^2z^2+7ixy^3+y^4+6y^2z^2-2z^4\right)}{84}$$
$$+\frac{5\sqrt{42}iG_z^{\alpha}\left(x-iy\right)^2\left(x^2+y^2-6z^2\right)}{168}-\frac{\sqrt{70}iG_z^{\beta}\left(5x^4-14ix^3y-18x^2y^2-12x^2z^2+14ixy^3+5y^4-12y^2z^2+4z^4\right)}{168}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{\sqrt{210}iG_{3c}z\left(x-iy\right)}{28}+\frac{\sqrt{210}G_{3s}z\left(x-iy\right)}{28}+\frac{3\sqrt{14}iG_{3x}z\left(x+iy\right)}{28}-\frac{3\sqrt{14}G_{3y}z\left(x+iy\right)}{28}$$
$$-\frac{\sqrt{35}G_{xyz}\left(x^2+y^2-2z^2\right)}{28}-\frac{\sqrt{21}iG_z^{\alpha}\left(x+iy\right)^2}{28}+\frac{\sqrt{35}iG_z^{\beta}\left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{\sqrt{105}iG_{3c}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{\sqrt{105}G_{3s}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{168}+\frac{5\sqrt{7}iG_{3x}z\left(x+iy\right)\left(13x^2+14ixy-y^2-8z^2\right)}{168}$$
$$+\frac{5\sqrt{7}G_{3y}z\left(x+iy\right)\left(x^2+14ixy-13y^2+8z^2\right)}{168}+\frac{\sqrt{70}G_{xyz}\left(x^4+7ix^3y-12x^2y^2+6x^2z^2-7ixy^3+y^4+6y^2z^2-2z^4\right)}{84}$$
$$-\frac{5\sqrt{42}iG_z^{\alpha}\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{168}+\frac{\sqrt{70}iG_z^{\beta}\left(5x^4+14ix^3y-18x^2y^2-12x^2z^2-14ixy^3+5y^4-12y^2z^2+4z^4\right)}{168}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{14}G_{3x}x}{7}+\frac{\sqrt{14}G_{3y}y}{7}+\frac{\sqrt{21}G_z^{\alpha}z}{7}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{5\sqrt{210}G_{3c}x\left(x^2-3y^2\right)}{168}-\frac{5\sqrt{210}G_{3s}y\left(3x^2-y^2\right)}{168}-\frac{3\sqrt{14}G_{3x}x\left(x^2+y^2-4z^2\right)}{56}-\frac{3\sqrt{14}G_{3y}y\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{21}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{1155}G_{3c}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{264}-\frac{\sqrt{1155}G_{3s}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{264}+\frac{15\sqrt{77}G_{3x}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{616}$$
$$+\frac{15\sqrt{77}G_{3y}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{616}-\frac{3\sqrt{770}G_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{5\sqrt{462}G_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}G_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{2\sqrt{21}G_{3x}z}{21}+\frac{2\sqrt{21}iG_{3y}z}{21}+\frac{\sqrt{210}iG_{xyz}\left(x-iy\right)}{42}-\frac{\sqrt{14}G_{z}^{\alpha}\left(x+iy\right)}{14}+\frac{\sqrt{210}G_{z}^{\beta}\left(x-iy\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{5\sqrt{35}G_{3c}z\left(x-iy\right)^{2}}{56}+\frac{5\sqrt{35}iG_{3s}z\left(x-iy\right)^{2}}{56}+\frac{\sqrt{21}G_{3x}z\left(9x^{2}+30ixy-21y^{2}+4z^{2}\right)}{168}-\frac{\sqrt{21}iG_{3y}z\left(21x^{2}+30ixy-9y^{2}-4z^{2}\right)}{168}$$
$$-\frac{\sqrt{210}iG_{xyz}\left(2x^{3}+3ix^{2}y-3xy^{2}-3xz^{2}-2iy^{3}+3iyz^{2}\right)}{84}-\frac{\sqrt{14}G_{z}^{\alpha}\left(x+iy\right)\left(x^{2}+y^{2}-4z^{2}\right)}{56}+\frac{\sqrt{210}G_{z}^{\beta}\left(x^{3}+9ix^{2}y-9xy^{2}+6xz^{2}-iy^{3}-6iyz^{2}\right)}{168}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{770}G_{3c}z\left(2x^{4}+5ix^{3}y-9x^{2}y^{2}-x^{2}z^{2}-7ixy^{3}+2ixyz^{2}+y^{4}+y^{2}z^{2}\right)}{88}$$
$$-\frac{\sqrt{770}iG_{3s}z\left(x^{4}+7ix^{3}y-9x^{2}y^{2}+x^{2}z^{2}-5ixy^{3}-2ixyz^{2}+2y^{4}-y^{2}z^{2}\right)}{88}$$
$$-\frac{5\sqrt{462}G_{3x}z\left(18x^{4}+21ix^{3}y+15x^{2}y^{2}-41x^{2}z^{2}+21ixy^{3}-42ixyz^{2}-3y^{4}+y^{2}z^{2}+4z^{4}\right)}{1848}$$
$$+\frac{5\sqrt{462}iG_{3y}z\left(3x^{4}+21ix^{3}y-15x^{2}y^{2}-x^{2}z^{2}+21ixy^{3}-42ixyz^{2}-18y^{4}+41y^{2}z^{2}-4z^{4}\right)}{1848}$$
$$+\frac{\sqrt{1155}iG_{xyz}\left(x^{5}+6ix^{4}y-5x^{3}y^{2}-5x^{3}z^{2}+5ix^{2}y^{3}-51ix^{2}yz^{2}-6xy^{4}+51xy^{2}z^{2}-6xz^{4}-iy^{5}+5iy^{3}z^{2}+6iyz^{4}\right)}{462}$$
$$+\frac{5\sqrt{77}G_{z}^{\alpha}\left(x+iy\right)\left(x^{4}+2x^{2}y^{2}-12x^{2}z^{2}+y^{4}-12y^{2}z^{2}+8z^{4}\right)}{308}$$
$$-\frac{\sqrt{1155}G_{z}^{\beta}\left(5x^{5}+9ix^{4}y-4x^{3}y^{2}-46x^{3}z^{2}+4ix^{2}y^{3}-66ix^{2}yz^{2}-9xy^{4}+66xy^{2}z^{2}+12xz^{4}-5iy^{5}+46iy^{3}z^{2}-12iyz^{4}\right)}{924}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{2\sqrt{21}G_{3x}z}{21}-\frac{2\sqrt{21}iG_{3y}z}{21}-\frac{\sqrt{210}iG_{xyz}\left(x+iy\right)}{42}-\frac{\sqrt{14}G_{z}^{\alpha}\left(x-iy\right)}{14}+\frac{\sqrt{210}G_{z}^{\beta}\left(x+iy\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{5\sqrt{35}G_{3c}z\left(x+iy\right)^{2}}{56}-\frac{5\sqrt{35}iG_{3s}z\left(x+iy\right)^{2}}{56}+\frac{\sqrt{21}G_{3x}z\left(9x^{2}-30ixy-21y^{2}+4z^{2}\right)}{168}+\frac{\sqrt{21}iG_{3y}z\left(21x^{2}-30ixy-9y^{2}-4z^{2}\right)}{168}$$
$$+\frac{\sqrt{210}iG_{xyz}\left(2x^{3}-3ix^{2}y-3xy^{2}-3xz^{2}+2iy^{3}-3iyz^{2}\right)}{84}-\frac{\sqrt{14}G_{z}^{\alpha}\left(x-iy\right)\left(x^{2}+y^{2}-4z^{2}\right)}{56}+\frac{\sqrt{210}G_{z}^{\beta}\left(x^{3}-9ix^{2}y-9xy^{2}+6xz^{2}+iy^{3}+6iyz^{2}\right)}{168}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{770}G_{3c}z\left(2x^4-5ix^3y-9x^2y^2-x^2z^2+7ixy^3-2ixyz^2+y^4+y^2z^2\right)}{88}$$
$$+\frac{\sqrt{770}iG_{3s}z\left(x^4-7ix^3y-9x^2y^2+x^2z^2+5ixy^3+2ixyz^2+2y^4-y^2z^2\right)}{88}$$
$$-\frac{5\sqrt{462}G_{3x}z\left(18x^4-21ix^3y+15x^2y^2-41x^2z^2-21ixy^3+42ixyz^2-3y^4+y^2z^2+4z^4\right)}{1848}$$
$$-\frac{5\sqrt{462}iG_{3y}z\left(3x^4-21ix^3y-15x^2y^2-x^2z^2-21ixy^3+42ixyz^2-18y^4+41y^2z^2-4z^4\right)}{1848}$$
$$-\frac{\sqrt{1155}iG_{xyz}\left(x^5-6ix^4y-5x^3y^2-5x^3z^2-5ix^2y^3+51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4+iy^5-5iy^3z^2-6iyz^4\right)}{462}$$
$$+\frac{5\sqrt{77}G_z^{\alpha}\left(x-iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}$$
$$-\frac{\sqrt{1155}G_z^{\beta}\left(5x^5-9ix^4y-4x^3y^2-46x^3z^2-4ix^2y^3+66ix^2yz^2-9xy^4+66xy^2z^2+12xz^4+5iy^5-46iy^3z^2+12iyz^4\right)}{924}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{35}G_{3c}\left(x+iy\right)}{14}-\frac{\sqrt{35}iG_{3s}\left(x+iy\right)}{14}-\frac{\sqrt{21}G_{3x}\left(x-iy\right)}{42}+\frac{\sqrt{21}iG_{3y}\left(x-iy\right)}{42}-\frac{\sqrt{210}iG_{xyz}z}{42}+\frac{\sqrt{210}G_z^{\beta}z}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{35}G_{3c}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{35}iG_{3s}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{21}G_{3x}\left(x-iy\right)\left(11x^2-10ixy+y^2-24z^2\right)}{168}$$
$$+\frac{\sqrt{21}iG_{3y}\left(x-iy\right)\left(x^2+10ixy+11y^2-24z^2\right)}{168}-\frac{\sqrt{210}iG_{xyz}z\left(3x^2+3y^2-2z^2\right)}{84}-\frac{5\sqrt{14}G_z^{\alpha}z\left(x-iy\right)^2}{28}+\frac{\sqrt{210}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{770}G_{3c}\left(11x^5-52ix^4y-104x^3y^2-6x^3z^2+106ix^2y^3-6ix^2yz^2+53xy^4-6xy^2z^2+4xz^4-10iy^5-6iy^3z^2+4iyz^4\right)}{1232}$$
$$+\frac{\sqrt{770}iG_{3s}\left(10x^5-53ix^4y-106x^3y^2+6x^3z^2+104ix^2y^3+6ix^2yz^2+52xy^4+6xy^2z^2-4xz^4-11iy^5+6iy^3z^2-4iyz^4\right)}{1232}$$
$$-\frac{5\sqrt{462}G_{3x}\left(x-iy\right)\left(5x^4-7ix^3y+3x^2y^2-46x^2z^2-7ixy^3+56ixyz^2-2y^4+10y^2z^2+12z^4\right)}{3696}$$
$$-\frac{5\sqrt{462}iG_{3y}\left(x-iy\right)\left(2x^4-7ix^3y-3x^2y^2-10x^2z^2-7ixy^3+56ixyz^2-5y^4+46y^2z^2-12z^4\right)}{3696}$$
$$+\frac{\sqrt{1155}iG_{xyz}z\left(12x^4-63ix^3y-102x^2y^2+10x^2z^2+63ixy^3+12y^4+10y^2z^2-2z^4\right)}{924}$$
$$-\frac{5\sqrt{77}G_z^{\alpha}z\left(x-iy\right)^2\left(x^2+y^2-2z^2\right)}{88}+\frac{\sqrt{1155}G_z^{\beta}z\left(39x^4-126ix^3y-174x^2y^2-20x^2z^2+126ixy^3+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](E_{2u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{35}G_{3c}\left(x-iy\right)}{14}+\frac{\sqrt{35}iG_{3s}\left(x-iy\right)}{14}-\frac{\sqrt{21}G_{3x}\left(x+iy\right)}{42}-\frac{\sqrt{21}iG_{3y}\left(x+iy\right)}{42}+\frac{\sqrt{210}iG_{xyz}z}{42}+\frac{\sqrt{210}G_z^{\beta}z}{42}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](E_{2u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{35}G_{3c}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{56}+\frac{\sqrt{35}iG_{3s}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{56}-\frac{\sqrt{21}G_{3x}\left(x+iy\right)\left(11x^2+10ixy+y^2-24z^2\right)}{168}$$
$$-\frac{\sqrt{21}iG_{3y}\left(x+iy\right)\left(x^2-10ixy+11y^2-24z^2\right)}{168}+\frac{\sqrt{210}iG_{xyz}z\left(3x^2+3y^2-2z^2\right)}{84}-\frac{5\sqrt{14}G_z^{\alpha}z\left(x+iy\right)^2}{28}+\frac{\sqrt{210}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{770}G_{3c}\left(11x^5+52ix^4y-104x^3y^2-6x^3z^2-106ix^2y^3+6ix^2yz^2+53xy^4-6xy^2z^2+4xz^4+10iy^5+6iy^3z^2-4iyz^4\right)}{1232}$$
$$-\frac{\sqrt{770}iG_{3s}\left(10x^5+53ix^4y-106x^3y^2+6x^3z^2-104ix^2y^3-6ix^2yz^2+52xy^4+6xy^2z^2-4xz^4+11iy^5-6iy^3z^2+4iyz^4\right)}{1232}$$
$$-\frac{5\sqrt{462}G_{3x}\left(x+iy\right)\left(5x^4+7ix^3y+3x^2y^2-46x^2z^2+7ixy^3-56ixyz^2-2y^4+10y^2z^2+12z^4\right)}{3696}$$
$$+\frac{5\sqrt{462}iG_{3y}\left(x+iy\right)\left(2x^4+7ix^3y-3x^2y^2-10x^2z^2+7ixy^3-56ixyz^2-5y^4+46y^2z^2-12z^4\right)}{3696}$$
$$-\frac{\sqrt{1155}iG_{xyz}z\left(12x^4+63ix^3y-102x^2y^2+10x^2z^2-63ixy^3+12y^4+10y^2z^2-2z^4\right)}{924}$$
$$-\frac{5\sqrt{77}G_z^{\alpha}z\left(x+iy\right)^2\left(x^2+y^2-2z^2\right)}{88}+\frac{\sqrt{1155}G_z^{\beta}z\left(39x^4+126ix^3y-174x^2y^2-20x^2z^2-126ixy^3+39y^4-20y^2z^2+4z^4\right)}{1848}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(3,-3)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$G_z^{\alpha}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{10}G_{3x}xz}{10}+\frac{\sqrt{10}G_{3y}yz}{10}-G_{xyz}xy-\frac{\sqrt{15}G_z^{\alpha}\left(x^2+y^2-2z^2\right)}{15}-\frac{G_z^{\beta}\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{7\sqrt{55}G_{3c}xz\left(x^2-3y^2\right)}{44}-\frac{7\sqrt{55}G_{3s}yz\left(3x^2-y^2\right)}{44}-\frac{5\sqrt{33}G_{3x}xz\left(3x^2+3y^2-4z^2\right)}{132}-\frac{5\sqrt{33}G_{3y}yz\left(3x^2+3y^2-4z^2\right)}{132}$$
$$+\frac{\sqrt{330}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{132}+\frac{\sqrt{22}G_z^{\alpha}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{88}+\frac{\sqrt{330}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{7\sqrt{4290}G_{3c}xz\left(x^2-3y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}-\frac{7\sqrt{4290}G_{3s}yz\left(3x^2-y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}$$
$$+\frac{35\sqrt{286}G_{3x}xz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}$$
$$+\frac{35\sqrt{286}G_{3y}yz\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}+\frac{7\sqrt{715}G_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}$$
$$-\frac{5\sqrt{429}G_z^{\alpha}\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

G_{3s}

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

G_{3c}

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{15}G_{3s}\left(x^2+y^2-2z^2\right)}{12}-\frac{G_{3x}xy}{2}-\frac{G_{3y}\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_{xyz}xz}{4}+\frac{\sqrt{10}G_z^{\beta}yz}{4}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{15}G_{3c}\left(x^2+y^2-2z^2\right)}{12}-\frac{G_{3x}\left(x-y\right)\left(x+y\right)}{4}+\frac{G_{3y}xy}{2}-\frac{\sqrt{10}G_{xyz}yz}{4}+\frac{\sqrt{10}G_z^{\beta}xz}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{22}G_{3s}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{176}-\frac{\sqrt{330}G_{3x}xy\left(5x^2-2y^2-9z^2\right)}{66}+\frac{\sqrt{330}G_{3y}\left(x^4-42x^2y^2+36x^2z^2+13y^4-36y^2z^2\right)}{528}$$
$$+\frac{5\sqrt{33}G_{xyz}xz\left(3x^2+3y^2-4z^2\right)}{132}-\frac{7\sqrt{55}G_z^{\alpha}yz\left(3x^2-y^2\right)}{44}+\frac{5\sqrt{33}G_z^{\beta}yz\left(3x^2+3y^2-4z^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{22}G_{3c}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{176}-\frac{\sqrt{330}G_{3x}\left(13x^4-42x^2y^2-36x^2z^2+y^4+36y^2z^2\right)}{528}$$
$$-\frac{\sqrt{330}G_{3y}xy\left(2x^2-5y^2+9z^2\right)}{66}-\frac{5\sqrt{33}G_{xyz}yz\left(3x^2+3y^2-4z^2\right)}{132}-\frac{7\sqrt{55}G_z^{\alpha}xz\left(x^2-3y^2\right)}{44}+\frac{5\sqrt{33}G_z^{\beta}xz\left(3x^2+3y^2-4z^2\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{7\sqrt{429}G_{3c}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{208}-\frac{\sqrt{429}G_{3s}\left(113x^6-1740x^4y^2+45x^4z^2+1725x^2y^4+90x^2y^2z^2-60x^2z^4-118y^6+45y^4z^2-60y^2z^4+8z^6\right)}{6864}$$
$$-\frac{7\sqrt{715}G_{3x}xy\left(7x^4+2x^2y^2-76x^2z^2-5y^4+44y^2z^2+16z^4\right)}{2288}$$
$$+\frac{7\sqrt{715}G_{3y}\left(x^6-8x^4y^2-7x^4z^2-7x^2y^4+90x^2y^2z^2-8x^2z^4+2y^6-23y^4z^2+8y^2z^4\right)}{2288}$$
$$-\frac{7\sqrt{286}G_{xyz}xz\left(7x^4-85x^2y^2+5x^2z^2+40y^4+5y^2z^2-2z^4\right)}{1144}-\frac{7\sqrt{4290}G_z^{\alpha}yz\left(3x^2-y^2\right)\left(3x^2+3y^2-8z^2\right)}{2288}$$
$$+\frac{7\sqrt{286}G_z^{\beta}yz\left(85x^4-160x^2y^2-10x^2z^2+19y^4-10y^2z^2+4z^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_g, 2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{429}G_{3c}\left(118x^6 - 1725x^4y^2 - 45x^4z^2 + 1740x^2y^4 - 90x^2y^2z^2 + 60x^2z^4 - 113y^6 - 45y^4z^2 + 60y^2z^4 - 8z^6\right)}{6864}$$
$$+ \frac{7\sqrt{429}G_{3s}xy\left(x^2 - 3y^2\right)\left(3x^2 - y^2\right)}{208} - \frac{7\sqrt{715}G_{3x}\left(2x^6 - 7x^4y^2 - 23x^4z^2 - 8x^2y^4 + 90x^2y^2z^2 + 8x^2z^4 + y^6 - 7y^4z^2 - 8y^2z^4\right)}{2288}$$
$$- \frac{7\sqrt{715}G_{3y}xy\left(5x^4 - 2x^2y^2 - 44x^2z^2 - 7y^4 + 76y^2z^2 - 16z^4\right)}{2288} + \frac{7\sqrt{286}G_{xyz}yz\left(40x^4 - 85x^2y^2 + 5x^2z^2 + 7y^4 + 5y^2z^2 - 2z^4\right)}{1144}$$
$$- \frac{7\sqrt{4290}G_z^{\alpha}xz\left(x^2 - 3y^2\right)\left(3x^2 + 3y^2 - 8z^2\right)}{2288} + \frac{7\sqrt{286}G_z^{\beta}xz\left(19x^4 - 160x^2y^2 - 10x^2z^2 + 85y^4 - 10y^2z^2 + 4z^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{2}G_{3x}}{2} + \frac{\sqrt{2}iG_{3y}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{2}G_{3c}\left(x - iy\right)^2}{8} - \frac{\sqrt{2}iG_{3s}\left(x - iy\right)^2}{8} + \frac{\sqrt{30}G_{3x}\left(x^2 + 4ixy - 3y^2 + 2z^2\right)}{40}$$
$$- \frac{\sqrt{30}iG_{3y}\left(3x^2 + 4ixy - y^2 - 2z^2\right)}{40} + \frac{\sqrt{3}iG_{xyz}z\left(x - iy\right)}{4} + \frac{\sqrt{5}G_z^{\alpha}z\left(x + iy\right)}{10} + \frac{\sqrt{3}G_z^{\beta}z\left(x - iy\right)}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{165}G_{3c}\left(13x^4 + 16ix^3y - 42x^2y^2 - 36x^2z^2 - 40ixy^3 + 72ixyz^2 + y^4 + 36y^2z^2\right)}{528}$$
$$+ \frac{\sqrt{165}iG_{3s}\left(x^4 + 40ix^3y - 42x^2y^2 + 36x^2z^2 - 16ixy^3 - 72ixyz^2 + 13y^4 - 36y^2z^2\right)}{528}$$
$$- \frac{\sqrt{11}G_{3x}\left(17x^4 + 40ix^3y - 6x^2y^2 - 96x^2z^2 + 40ixy^3 - 240ixyz^2 - 23y^4 + 144y^2z^2 - 8z^4\right)}{528}$$
$$+ \frac{\sqrt{11}iG_{3y}\left(23x^4 + 40ix^3y + 6x^2y^2 - 144x^2z^2 + 40ixy^3 - 240ixyz^2 - 17y^4 + 96y^2z^2 + 8z^4\right)}{528}$$
$$- \frac{\sqrt{110}iG_{xyz}z\left(19x^3 + 9ix^2y - 9xy^2 - 16xz^2 - 19iy^3 + 16iyz^2\right)}{264}$$
$$- \frac{5\sqrt{66}G_z^{\alpha}z\left(x + iy\right)\left(3x^2 + 3y^2 - 4z^2\right)}{264} - \frac{\sqrt{110}G_z^{\beta}z\left(5x^3 - 33ix^2y + 33xy^2 - 16xz^2 - 5iy^3 + 16iyz^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{7\sqrt{1430}G_{3c}\left(2x^6+5ix^5y-7x^4y^2-23x^4z^2-2ix^3y^3-44ix^3yz^2-8x^2y^4+90x^2y^2z^2+8x^2z^4-7ixy^5+76ixy^3z^2-16ixyz^4+y^6-7y^4z^2-8y^2z^4\right)}{4576}$$
$$+\frac{7\sqrt{1430}iG_{3s}\left(x^6+7ix^5y-8x^4y^2-7x^4z^2+2ix^3y^3-76ix^3yz^2-7x^2y^4+90x^2y^2z^2-8x^2z^4-5ixy^5+44ixy^3z^2+16ixyz^4+2y^6-23y^4z^2+8y^2z^4\right)}{4576}$$
$$+\frac{5\sqrt{858}G_{3x}\left(6x^6+7ix^5y+11x^4y^2-101x^4z^2+14ix^3y^3-112ix^3yz^2+4x^2y^4-90x^2y^2z^2+116x^2z^4+7ixy^5-112ixy^3z^2+112ixyz^4-y^6+11y^4z^2+4y^2z^4\right)}{4576}$$
$$-\frac{5\sqrt{858}iG_{3y}\left(x^6+7ix^5y-4x^4y^2-11x^4z^2+14ix^3y^3-112ix^3yz^2-11x^2y^4+90x^2y^2z^2-4x^2z^4+7ixy^5-112ixy^3z^2+112ixyz^4-6y^6+101y^4z^2-116y^2z^4\right)}{4576}$$
$$+\frac{7\sqrt{2145}iG_{xyz}z\left(x^5+8ix^4y-7x^3y^2-x^3z^2+7ix^2y^3-23ix^2yz^2-8xy^4+23xy^2z^2-2xz^4-iy^5+iy^3z^2+2iyz^4\right)}{1144}$$
$$+\frac{35\sqrt{143}G_z^{\alpha}z\left(x+iy\right)\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}$$
$$-\frac{7\sqrt{2145}G_z^{\beta}z\left(7x^5+11ix^4y-4x^3y^2-22x^3z^2+4ix^2y^3-26ix^2yz^2-11xy^4+26xy^2z^2+4xz^4-7iy^5+22iy^3z^2-4iyz^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{2}G_{3x}}{2}-\frac{\sqrt{2}iG_{3y}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{2}G_{3c}\left(x+iy\right)^2}{8}+\frac{\sqrt{2}iG_{3s}\left(x+iy\right)^2}{8}+\frac{\sqrt{30}G_{3x}\left(x^2-4ixy-3y^2+2z^2\right)}{40}$$
$$+\frac{\sqrt{30}iG_{3y}\left(3x^2-4ixy-y^2-2z^2\right)}{40}-\frac{\sqrt{3}iG_{xyz}z\left(x+iy\right)}{4}+\frac{\sqrt{5}G_z^{\alpha}z\left(x-iy\right)}{10}+\frac{\sqrt{3}G_z^{\beta}z\left(x+iy\right)}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{165}G_{3c}\left(13x^4-16ix^3y-42x^2y^2-36x^2z^2+40ixy^3-72ixyz^2+y^4+36y^2z^2\right)}{528}$$
$$-\frac{\sqrt{165}iG_{3s}\left(x^4-40ix^3y-42x^2y^2+36x^2z^2+16ixy^3+72ixyz^2+13y^4-36y^2z^2\right)}{528}$$
$$-\frac{\sqrt{11}G_{3x}\left(17x^4-40ix^3y-6x^2y^2-96x^2z^2-40ixy^3+240ixyz^2-23y^4+144y^2z^2-8z^4\right)}{528}$$
$$-\frac{\sqrt{11}iG_{3y}\left(23x^4-40ix^3y+6x^2y^2-144x^2z^2-40ixy^3+240ixyz^2-17y^4+96y^2z^2+8z^4\right)}{528}$$
$$+\frac{\sqrt{110}iG_{xyz}z\left(19x^3-9ix^2y-9xy^2-16xz^2+19iy^3-16iyz^2\right)}{264}$$
$$-\frac{5\sqrt{66}G_z^{\alpha}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{264}-\frac{\sqrt{110}G_z^{\beta}z\left(5x^3+33ix^2y+33xy^2-16xz^2+5iy^3-16iyz^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{7\sqrt{1430}G_{3c}\left(2x^6-5ix^5y-7x^4y^2-23x^4z^2+2ix^3y^3+44ix^3yz^2-8x^2y^4+90x^2y^2z^2+8x^2z^4+7ixy^5-76ixy^3z^2+16ixyz^4+y^6-7y^4z^2-8y^2z^4\right)}{4576}$$
$$-\frac{7\sqrt{1430}iG_{3s}\left(x^6-7ix^5y-8x^4y^2-7x^4z^2-2ix^3y^3+76ix^3yz^2-7x^2y^4+90x^2y^2z^2-8x^2z^4+5ixy^5-44ixy^3z^2-16ixyz^4+2y^6-23y^4z^2+8y^2z^4\right)}{4576}$$
$$+\frac{5\sqrt{858}G_{3x}\left(6x^6-7ix^5y+11x^4y^2-101x^4z^2-14ix^3y^3+112ix^3yz^2+4x^2y^4-90x^2y^2z^2+116x^2z^4-7ixy^5+112ixy^3z^2-112ixyz^4-y^6+11y^4z^2+4y^2z^4\right.}{4576}$$
$$+\frac{5\sqrt{858}iG_{3y}\left(x^6-7ix^5y-4x^4y^2-11x^4z^2-14ix^3y^3+112ix^3yz^2-11x^2y^4+90x^2y^2z^2-4x^2z^4-7ixy^5+112ixy^3z^2-112ixyz^4-6y^6+101y^4z^2-116y^2z^4\right.}{4576}$$
$$-\frac{7\sqrt{2145}iG_{xyz}z\left(x^5-8ix^4y-7x^3y^2-x^3z^2-7ix^2y^3+23ix^2yz^2-8xy^4+23xy^2z^2-2xz^4+iy^5-iy^3z^2-2iyz^4\right)}{1144}$$
$$+\frac{35\sqrt{143}G_z^{\alpha}z\left(x-iy\right)\left(5x^4+10x^2y^2-20x^2z^2+5y^4-20y^2z^2+8z^4\right)}{2288}$$
$$-\frac{7\sqrt{2145}G_z^{\beta}z\left(7x^5-11ix^4y-4x^3y^2-22x^3z^2-4ix^2y^3+26ix^2yz^2-11xy^4+26xy^2z^2+4xz^4+7iy^5-22iy^3z^2+4iyz^4\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{\sqrt{2}iG_{xyz}}{2}+\frac{\sqrt{2}G_z^{\beta}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{5}G_{3c}z\left(x+iy\right)}{4}-\frac{\sqrt{5}iG_{3s}z\left(x+iy\right)}{4}+\frac{\sqrt{3}G_{3x}z\left(x-iy\right)}{4}-\frac{\sqrt{3}iG_{3y}z\left(x-iy\right)}{4}-\frac{\sqrt{2}G_z^{\alpha}\left(x-iy\right)^2}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{5\sqrt{66}G_{3c}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{264}-\frac{5\sqrt{66}iG_{3s}z\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{264}-\frac{\sqrt{110}G_{3x}z\left(x-iy\right)\left(5x^2+14ixy+19y^2-16z^2\right)}{264}$$
$$+\frac{\sqrt{110}iG_{3y}z\left(x-iy\right)\left(19x^2-14ixy+5y^2-16z^2\right)}{264}+\frac{7\sqrt{11}iG_{xyz}\left(2x^4-5ix^3y-6x^2y^2-6x^2z^2+5ixy^3+2y^4-6y^2z^2+2z^4\right)}{132}$$
$$+\frac{\sqrt{165}G_z^{\alpha}\left(x-iy\right)^2\left(x^2+y^2-6z^2\right)}{264}+\frac{7\sqrt{11}G_z^{\beta}\left(x^4-10ix^3y-18x^2y^2+12x^2z^2+10ixy^3+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{7\sqrt{143}G_{3c}z\left(19x^5-80ix^4y-160x^3y^2-10x^3z^2+170ix^2y^3-10ix^2yz^2+85xy^4-10xy^2z^2+4xz^4-14iy^5-10iy^3z^2+4iyz^4\right)}{2288}$$
$$+\frac{7\sqrt{143}iG_{3s}z\left(14x^5-85ix^4y-170x^3y^2+10x^3z^2+160ix^2y^3+10ix^2yz^2+80xy^4+10xy^2z^2-4xz^4-19iy^5+10iy^3z^2-4iyz^4\right)}{2288}$$
$$-\frac{7\sqrt{2145}G_{3x}z\left(x-iy\right)\left(7x^4-9ix^3y+5x^2y^2-22x^2z^2-9ixy^3+24ixyz^2-2y^4+2y^2z^2+4z^4\right)}{2288}$$
$$-\frac{7\sqrt{2145}iG_{3y}z\left(x-iy\right)\left(2x^4-9ix^3y-5x^2y^2-2x^2z^2-9ixy^3+24ixyz^2-7y^4+22y^2z^2-4z^4\right)}{2288}$$
$$-\frac{\sqrt{858}iG_{xyz}\left(4x^6-21ix^5y-30x^4y^2-30x^4z^2+210ix^3yz^2-30x^2y^4+360x^2y^2z^2-30x^2z^4+21ixy^5-210ixy^3z^2+4y^6-30y^4z^2-30y^2z^4+4z^6\right)}{1144}$$
$$+\frac{7\sqrt{1430}G_z^{\alpha}\left(x-iy\right)^2\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{2288}$$
$$-\frac{\sqrt{858}G_z^{\beta}\left(13x^6-42ix^5y-45x^4y^2-150x^4z^2+420ix^3yz^2-45x^2y^4+540x^2y^2z^2+60x^2z^4+42ixy^5-420ixy^3z^2+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{2288}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}iG_{xyz}}{2}+\frac{\sqrt{2}G_z^{\beta}}{2}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{5}G_{3c}z\left(x-iy\right)}{4}+\frac{\sqrt{5}iG_{3s}z\left(x-iy\right)}{4}+\frac{\sqrt{3}G_{3x}z\left(x+iy\right)}{4}+\frac{\sqrt{3}iG_{3y}z\left(x+iy\right)}{4}-\frac{\sqrt{2}G_z^{\alpha}\left(x+iy\right)^2}{4}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{5\sqrt{66}G_{3c}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{264}+\frac{5\sqrt{66}iG_{3s}z\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{264}-\frac{\sqrt{110}G_{3x}z\left(x+iy\right)\left(5x^2-14ixy+19y^2-16z^2\right)}{264}$$
$$-\frac{\sqrt{110}iG_{3y}z\left(x+iy\right)\left(19x^2+14ixy+5y^2-16z^2\right)}{264}-\frac{7\sqrt{11}iG_{xyz}\left(2x^4+5ix^3y-6x^2y^2-6x^2z^2-5ixy^3+2y^4-6y^2z^2+2z^4\right)}{132}$$
$$+\frac{\sqrt{165}G_z^{\alpha}\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{264}+\frac{7\sqrt{11}G_z^{\beta}\left(x^4+10ix^3y-18x^2y^2+12x^2z^2-10ixy^3+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{7\sqrt{143}G_{3c}z\left(19x^5+80ix^4y-160x^3y^2-10x^3z^2-170ix^2y^3+10ix^2yz^2+85xy^4-10xy^2z^2+4xz^4+14iy^5+10iy^3z^2-4iyz^4\right)}{2288}$$
$$-\frac{7\sqrt{143}iG_{3s}z\left(14x^5+85ix^4y-170x^3y^2+10x^3z^2-160ix^2y^3-10ix^2yz^2+80xy^4+10xy^2z^2-4xz^4+19iy^5-10iy^3z^2+4iyz^4\right)}{2288}$$
$$-\frac{7\sqrt{2145}G_{3x}z\left(x+iy\right)\left(7x^4+9ix^3y+5x^2y^2-22x^2z^2+9ixy^3-24ixyz^2-2y^4+2y^2z^2+4z^4\right)}{2288}$$
$$+\frac{7\sqrt{2145}iG_{3y}z\left(x+iy\right)\left(2x^4+9ix^3y-5x^2y^2-2x^2z^2+9ixy^3-24ixyz^2-7y^4+22y^2z^2-4z^4\right)}{2288}$$
$$+\frac{\sqrt{858}iG_{xyz}\left(4x^6+21ix^5y-30x^4y^2-30x^4z^2-210ix^3yz^2-30x^2y^4+360x^2y^2z^2-30x^2z^4-21ixy^5+210ixy^3z^2+4y^6-30y^4z^2-30y^2z^4+4z^6\right)}{1144}$$
$$+\frac{7\sqrt{1430}G_z^{\alpha}\left(x+iy\right)^2\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{2288}$$
$$-\frac{\sqrt{858}G_z^{\beta}\left(13x^6+42ix^5y-45x^4y^2-150x^4z^2-420ix^3yz^2-45x^2y^4+540x^2y^2z^2+60x^2z^4-42ixy^5+420ixy^3z^2+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{2288}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](A_u)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{2}G_{3x}y}{2}-\frac{\sqrt{2}G_{3y}x}{2}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](A_u)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_{3c}y\left(3x^2-y^2\right)}{12}+\frac{\sqrt{15}G_{3s}x\left(x^2-3y^2\right)}{12}-\frac{G_{3x}y\left(x^2+y^2-4z^2\right)}{4}+\frac{G_{3y}x\left(x^2+y^2-4z^2\right)}{4}-\frac{\sqrt{10}G_{xyz}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_z^{\beta}xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](A_u)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{7\sqrt{330}G_{3c}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{7\sqrt{330}G_{3s}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{5\sqrt{22}G_{3x}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}$$
$$-\frac{5\sqrt{22}G_{3y}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}+\frac{7\sqrt{55}G_{xyz}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}-\frac{7\sqrt{55}G_z^{\beta}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_u,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{3}G_{3c}z}{2}-\frac{\sqrt{2}G_{xyz}y}{4}+\frac{\sqrt{2}G_z^{\beta}x}{4}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_u,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{3}G_{3s}z}{2}-\frac{\sqrt{2}G_{xyz}x}{4}-\frac{\sqrt{2}G_z^{\beta}y}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_u,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{6}G_{3c}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_{3x}z\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{10}G_{3y}xyz}{2}-\frac{G_{xyz}y\left(x^2+y^2-4z^2\right)}{4}-\frac{\sqrt{15}G_z^{\alpha}x\left(x^2-3y^2\right)}{12}+\frac{G_z^{\beta}x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_u,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{6}G_{3s}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{10}G_{3x}xyz}{2}-\frac{\sqrt{10}G_{3y}z\left(x-y\right)\left(x+y\right)}{4}-\frac{G_{xyz}x\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{15}G_z^{\alpha}y\left(3x^2-y^2\right)}{12}-\frac{G_z^{\beta}y\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_u,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{33}G_{3c}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528}+\frac{7\sqrt{55}G_{3x}z\left(5x^4-18x^2y^2-4x^2z^2+y^4+4y^2z^2\right)}{176}$$
$$+\frac{7\sqrt{55}G_{3y}xyz\left(x^2-2y^2+z^2\right)}{22}+\frac{\sqrt{22}G_{xyz}y\left(25x^4-55x^2y^2+15x^2z^2+4y^4+15y^2z^2-10z^4\right)}{88}$$
$$-\frac{7\sqrt{330}G_z^{\alpha}x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{528}+\frac{\sqrt{22}G_z^{\beta}x\left(13x^4-100x^2y^2-30x^2z^2+55y^4-30y^2z^2+20z^4\right)}{176}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_u,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{33}G_{3s}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{528}-\frac{7\sqrt{55}G_{3x}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$+\frac{7\sqrt{55}G_{3y}z\left(x^4-18x^2y^2+4x^2z^2+5y^4-4y^2z^2\right)}{176}+\frac{\sqrt{22}G_{xyz}x\left(4x^4-55x^2y^2+15x^2z^2+25y^4+15y^2z^2-10z^4\right)}{88}$$
$$+\frac{7\sqrt{330}G_z^{\alpha}y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{528}-\frac{\sqrt{22}G_z^{\beta}y\left(55x^4-100x^2y^2-30x^2z^2+13y^4-30y^2z^2+20z^4\right)}{176}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{6}iG_{3x}z}{12}+\frac{\sqrt{6}G_{3y}z}{12}-\frac{\sqrt{15}G_{xyz}\left(x-iy\right)}{12}+\frac{iG_z^{\alpha}\left(x+iy\right)}{2}+\frac{\sqrt{15}iG_z^{\beta}\left(x-iy\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{5}iG_{3c}z\left(x-iy\right)^2}{4}-\frac{\sqrt{5}G_{3s}z\left(x-iy\right)^2}{4}+\frac{\sqrt{3}iG_{3x}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{3}G_{3y}z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{30}G_{xyz}\left(x+iy\right)^3}{24}-\frac{\sqrt{2}iG_z^{\alpha}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{8}-\frac{\sqrt{30}iG_z^{\beta}\left(x+iy\right)^3}{24}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$-\frac{7\sqrt{110}iG_{3c}z\left(x^4+16ix^3y-18x^2y^2+4x^2z^2-8ixy^3-8ixyz^2+5y^4-4y^2z^2\right)}{352}$$
$$-\frac{7\sqrt{110}G_{3s}z\left(5x^4+8ix^3y-18x^2y^2-4x^2z^2-16ixy^3+8ixyz^2+y^4+4y^2z^2\right)}{352}$$
$$-\frac{5\sqrt{66}iG_{3x}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1056}+\frac{5\sqrt{66}G_{3y}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1056}$$
$$+\frac{\sqrt{165}G_{xyz}\left(4x^5+3ix^4y+x^3y^2-41x^3z^2-ix^2y^3-15ix^2yz^2-3xy^4+15xy^2z^2+18xz^4-4iy^5+41iy^3z^2-18iyz^4\right)}{264}$$
$$+\frac{5\sqrt{11}iG_z^{\alpha}\left(x+iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}$$
$$-\frac{\sqrt{165}iG_z^{\beta}\left(x^5-15ix^4y+16x^3y^2-26x^3z^2-16ix^2y^3+138ix^2yz^2+15xy^4-138xy^2z^2+36xz^4-iy^5+26iy^3z^2-36iyz^4\right)}{528}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E_{1u}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{6}iG_{3x}z}{12}+\frac{\sqrt{6}G_{3y}z}{12}-\frac{\sqrt{15}G_{xyz}\left(x+iy\right)}{12}-\frac{iG_z^{\alpha}\left(x-iy\right)}{2}-\frac{\sqrt{15}iG_z^{\beta}\left(x+iy\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E_{1u}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{5}iG_{3c}z\left(x+iy\right)^2}{4}-\frac{\sqrt{5}G_{3s}z\left(x+iy\right)^2}{4}-\frac{\sqrt{3}iG_{3x}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{3}G_{3y}z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{30}G_{xyz}\left(x-iy\right)^3}{24}+\frac{\sqrt{2}iG_z^{\alpha}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{8}+\frac{\sqrt{30}iG_z^{\beta}\left(x-iy\right)^3}{24}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E_{1u}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\begin{aligned}&\frac{7\sqrt{110}iG_{3c}z\left(x^4-16ix^3y-18x^2y^2+4x^2z^2+8ixy^3+8ixyz^2+5y^4-4y^2z^2\right)}{352}\\&-\frac{7\sqrt{110}G_{3s}z\left(5x^4-8ix^3y-18x^2y^2-4x^2z^2+16ixy^3-8ixyz^2+y^4+4y^2z^2\right)}{352}\\&+\frac{5\sqrt{66}iG_{3x}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1056}+\frac{5\sqrt{66}G_{3y}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1056}\\&+\frac{\sqrt{165}G_{xyz}\left(4x^5-3ix^4y+x^3y^2-41x^3z^2+ix^2y^3+15ix^2yz^2-3xy^4+15xy^2z^2+18xz^4+4iy^5-41iy^3z^2+18iyz^4\right)}{264}\\&-\frac{5\sqrt{11}iG_z^{\alpha}\left(x-iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}\\&+\frac{\sqrt{165}iG_z^{\beta}\left(x^5+15ix^4y+16x^3y^2-26x^3z^2+16ix^2y^3-138ix^2yz^2+15xy^4-138xy^2z^2+36xz^4+iy^5-26iy^3z^2+36iyz^4\right)}{528}\end{aligned}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{iG_{3c}\left(x+iy\right)}{4}-\frac{G_{3s}\left(x+iy\right)}{4}-\frac{\sqrt{15}iG_{3x}\left(x-iy\right)}{12}-\frac{\sqrt{15}G_{3y}\left(x-iy\right)}{12}+\frac{\sqrt{6}G_{xyz}z}{6}+\frac{\sqrt{6}iG_z^{\beta}z}{6}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\begin{aligned}&-\frac{\sqrt{2}iG_{3c}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{8}-\frac{\sqrt{2}G_{3s}\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{8}-\frac{\sqrt{30}iG_{3x}\left(x-iy\right)^3}{24}\\&+\frac{\sqrt{30}G_{3y}\left(x-iy\right)^3}{24}-\frac{\sqrt{3}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{5}iG_z^{\alpha}z\left(x-iy\right)^2}{4}-\frac{\sqrt{3}iG_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}\end{aligned}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\begin{aligned}&\frac{\sqrt{11}iG_{3c}\left(8x^5-55ix^4y-110x^3y^2+30x^3z^2+100ix^2y^3+30ix^2yz^2+50xy^4+30xy^2z^2-20xz^4-13iy^5+30iy^3z^2-20iyz^4\right)}{176}\\&-\frac{\sqrt{11}G_{3s}\left(13x^5-50ix^4y-100x^3y^2-30x^3z^2+110ix^2y^3-30ix^2yz^2+55xy^4-30xy^2z^2+20xz^4-8iy^5-30iy^3z^2+20iyz^4\right)}{176}\\&+\frac{\sqrt{165}iG_{3x}\left(x-iy\right)\left(8x^4-7ix^3y+9x^2y^2-82x^2z^2-7ixy^3+56ixyz^2+y^4-26y^2z^2+36z^4\right)}{528}\\&+\frac{\sqrt{165}G_{3y}\left(x-iy\right)\left(x^4+7ix^3y+9x^2y^2-26x^2z^2+7ixy^3-56ixyz^2+8y^4-82y^2z^2+36z^4\right)}{528}\\&-\frac{\sqrt{66}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}\\&+\frac{7\sqrt{110}iG_z^{\alpha}z\left(x-iy\right)^2\left(x^2+y^2-2z^2\right)}{88}-\frac{\sqrt{66}iG_z^{\beta}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}\end{aligned}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{iG_{3c}\left(x-iy\right)}{4}-\frac{G_{3s}\left(x-iy\right)}{4}+\frac{\sqrt{15}iG_{3x}\left(x+iy\right)}{12}-\frac{\sqrt{15}G_{3y}\left(x+iy\right)}{12}+\frac{\sqrt{6}G_{xyz}z}{6}-\frac{\sqrt{6}iG_z^{\beta}z}{6}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}iG_{3c}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{8}-\frac{\sqrt{2}G_{3s}\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{8}+\frac{\sqrt{30}iG_{3x}\left(x+iy\right)^3}{24}$$
$$+\frac{\sqrt{30}G_{3y}\left(x+iy\right)^3}{24}-\frac{\sqrt{3}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{5}iG_z^{\alpha}z\left(x+iy\right)^2}{4}+\frac{\sqrt{3}iG_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{\sqrt{11}iG_{3c}\left(8x^5+55ix^4y-110x^3y^2+30x^3z^2-100ix^2y^3-30ix^2yz^2+50xy^4+30xy^2z^2-20xz^4+13iy^5-30iy^3z^2+20iyz^4\right)}{176}$$
$$-\frac{\sqrt{11}G_{3s}\left(13x^5+50ix^4y-100x^3y^2-30x^3z^2-110ix^2y^3+30ix^2yz^2+55xy^4-30xy^2z^2+20xz^4+8iy^5+30iy^3z^2-20iyz^4\right)}{176}$$
$$-\frac{\sqrt{165}iG_{3x}\left(x+iy\right)\left(8x^4+7ix^3y+9x^2y^2-82x^2z^2+7ixy^3-56ixyz^2+y^4-26y^2z^2+36z^4\right)}{528}$$
$$+\frac{\sqrt{165}G_{3y}\left(x+iy\right)\left(x^4-7ix^3y+9x^2y^2-26x^2z^2-7ixy^3+56ixyz^2+8y^4-82y^2z^2+36z^4\right)}{528}$$
$$-\frac{\sqrt{66}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}$$
$$-\frac{7\sqrt{110}iG_z^{\alpha}z\left(x+iy\right)^2\left(x^2+y^2-2z^2\right)}{88}+\frac{\sqrt{66}iG_z^{\beta}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}$$